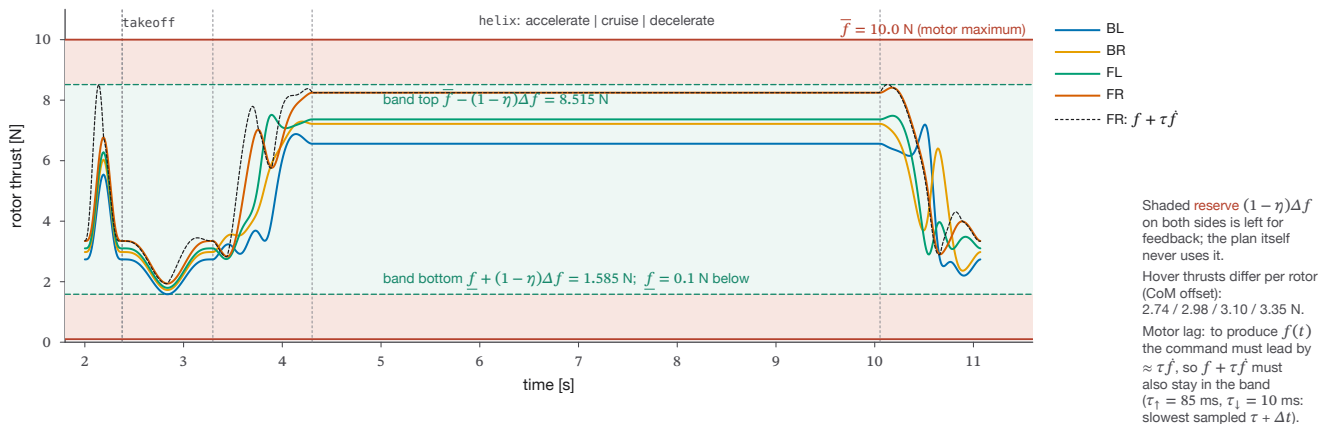
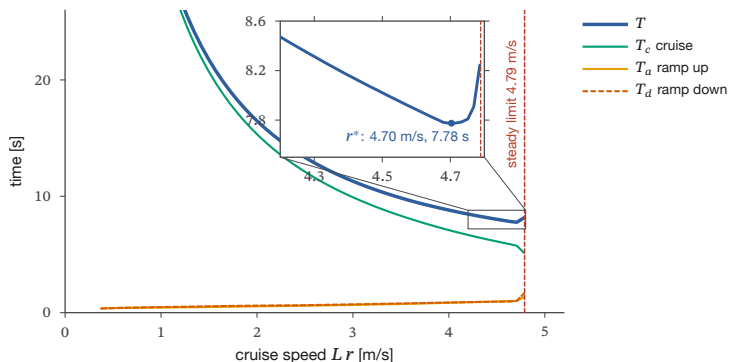


a Nominal rotor thrust of the planned motion must stay in the planning band ideal tracking, $\eta = \text{thrust_utilization} = 0.85$



b

helix phase: total time $T(r) = T_a + T_c + T_d$



Faster cruise shortens T_c but needs longer ramps; near the steady limit (centripetal thrust at the band top) the ramps diverge. Yaw-rate cap $Lr \leq 6.36 \text{ m/s}$ is not active. Takeoff: $T^* = 1.30 \text{ s}$, peak 1.54 m/s (no steady limit).

c

Planner trajectories/min_time.py, per null-duration segment

Check at every sample (`_PhaseLimits.feasible`): nominal wrench $F = m(a_d - g)$, $M = J\Omega_d + \Omega_d \times J\Omega_d$, unique 4-rotor solve $f = A_{[z,M]}^{-1}(\|F\|, M)$; require f and $f + \tau \dot{f}$ in the band, $\text{tilt} \leq \hat{\theta}_{\max}$, $\|a_d\| \leq a_{\max}$.

1 Steady limit: $\hat{\sigma} \approx r$ along the whole path (≥ 16 samples per turn); double r until infeasible, then bisection $\Rightarrow r_{\max}$.

2 Shortest ramps for a given r : 33 samples per ramp, geometric search then bisection $\Rightarrow T_a(r)$, $T_d(r)$ (ramps depend only on r).

3 Golden-section search of $T(r)$ on $(0.02r_{\max}, r_{\max}]$; $T_c < 0$ counts as infeasible. $\Rightarrow \text{CruiseTimeLaw}(r^*, T_a, T_d)$.

4 Re-check with `validate_trajectory_feasibility`; the report states `continuous_time_certificate: false, closed_loop_guarantee: false`.